

Explainable Multi-Stage Deep Learning for Brain Tumor Segmentation and Classification

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The project's goal is to investigate automated classification of brain tumor types on MRI scans using deep learning-based computer vision techniques, which will be combined with report generation for improved interpretability, addressing the critical challenge of accurate and timely diagnosis in neuro-oncology. This issue is particularly intriguing and important because brain tumors vary greatly in malignancy and morphology, resulting in high misdiagnosis rates with traditional radiological assessments, delaying life-saving interventions [1]. To provide context and background, we will review key literature on brain tumor classification with Magnetic Resonance Imaging (MRI), vision-language models (VLMs) for medical imaging, and automated report generation. This includes foundational works on VLMs and transformers, benchmarking studies for brain cancer diagnosis, and hybrid approaches integrating CNNs with VLMs for enhanced interpretability and clinical relevance. Additionally, we will also examine a cohort study on the accuracy of conventional MRI for preoperative diagnosis of intracranial tumours, foundational papers on ResNet50, InceptionV3, and recent publications on optimised deep learning architectures for brain tumour classification, such as those employing the improved Hunger Games Search Algorithm [2]. The data we will use is the publicly available Brain Tumor MRI Dataset from kaggle, which contains human brain MRI images across four classes: “Glioma”, “Meningioma”, “Pituitary tumour”, and “No tumour” with no new data collection required. Also, the SciDB Brain Tumor Dataset provides MRI scans with tumor segmentation masks and pixel-level annotations. For the method, we propose a multi-stage pipeline starting with a segmentation model based on U-Net or nnU-Net to isolate tumor regions from input, reducing noise and focusing on key features. The segmented outputs feed into a fine-tuned CNN classifier like ResNet50 or InceptionV3, using transfer learning for predicting tumor types. For explainability, an AI module applies Grad-CAM for visual heatmaps and a VLM like CLIP to generate reports on rationale, characteristics, and confidence. We will use existing implementations, modifying them by adding segmentation preprocessing for robustness, custom VLM fine-tuning for medical reports, and hyperparameter optimization via grid search to handle imbalances. Exact details will evolve based on tests. The evaluation will adopt both qualitative and quantitative methods. Qualitatively, segmentation overlays highlight agreements and differences with ground truth, while Grad-CAM focuses on actual tumor regions. Additionally, confusion matrices illustrate the distribution. Quantitatively, for segmentation, metrics will include the Dice Similarity Coefficient, Hausdorff Distance, Sensitivity, and Specificity. For classification, the model will be assessed using accuracy, precision and recall, and F1-score.

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References

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